Big Data Challenges and Insights

Exploring Tools and Techniques

By Room 3

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IMDb dataset

IMDb Dataset

Dataset from IMDb to make a recommendation system

Data Card | Code (48) | Discussion (5) | Suggestions (0)

About Dataset

Context

Each dataset is contained in a gzipped, tab-separated-values (TSV) formatted file in the UTF-8 character set. The first line in each file contains headers that describe what is in each column. A '\N' is used to denote that a particular field is missing or null for that title/name. The available datasets are as follows:

Content

title.akas.tsv.gz - Contains the following information for titles:

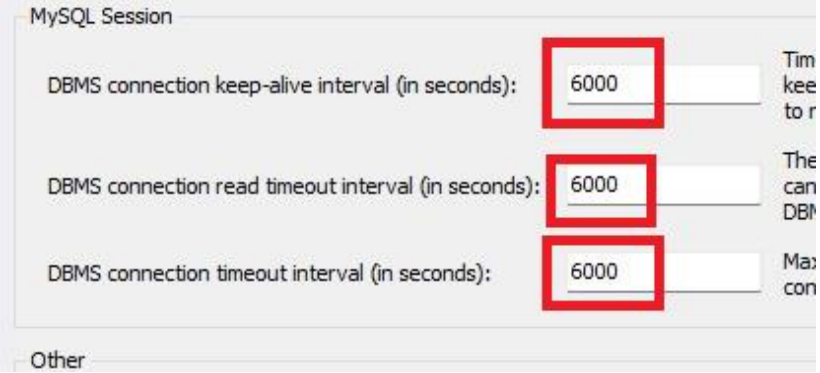
- titleId (string) - a tconst, an alphanumeric unique identifier of the title.
- ordering (integer) – a number to uniquely identify rows for a given titleId.
- title (string) – the localized title.
- region (string) - the region for this version of the title.
- language (string) - the language of the title.
- types (array) - Enumerated set of attributes for this alternative title. One or more of the following: "alternative", "dvd", "festival", "tv", "video", "working", "original", "imdbDisplay". New values may be added in the future without warning.

Large data files. Several 10s of millions of records – several interlinked data files

Understanding Big Data Challenges

- 1. Managing large file sizes.
- 2. Splitting and processing data efficiently.
- 3. Utilizing tools for optimal performance.
- 4. Monitoring and debugging using various utilities.

Setting mySQL workbench preferences



This increases the time mySQL workbench will wait until reporting an error when waiting for the mySQL server to do long operations.

Without this you will keep getting the following message from mySQL Workbench.

Error Code: 2013. Lost connection to MySQL server during query

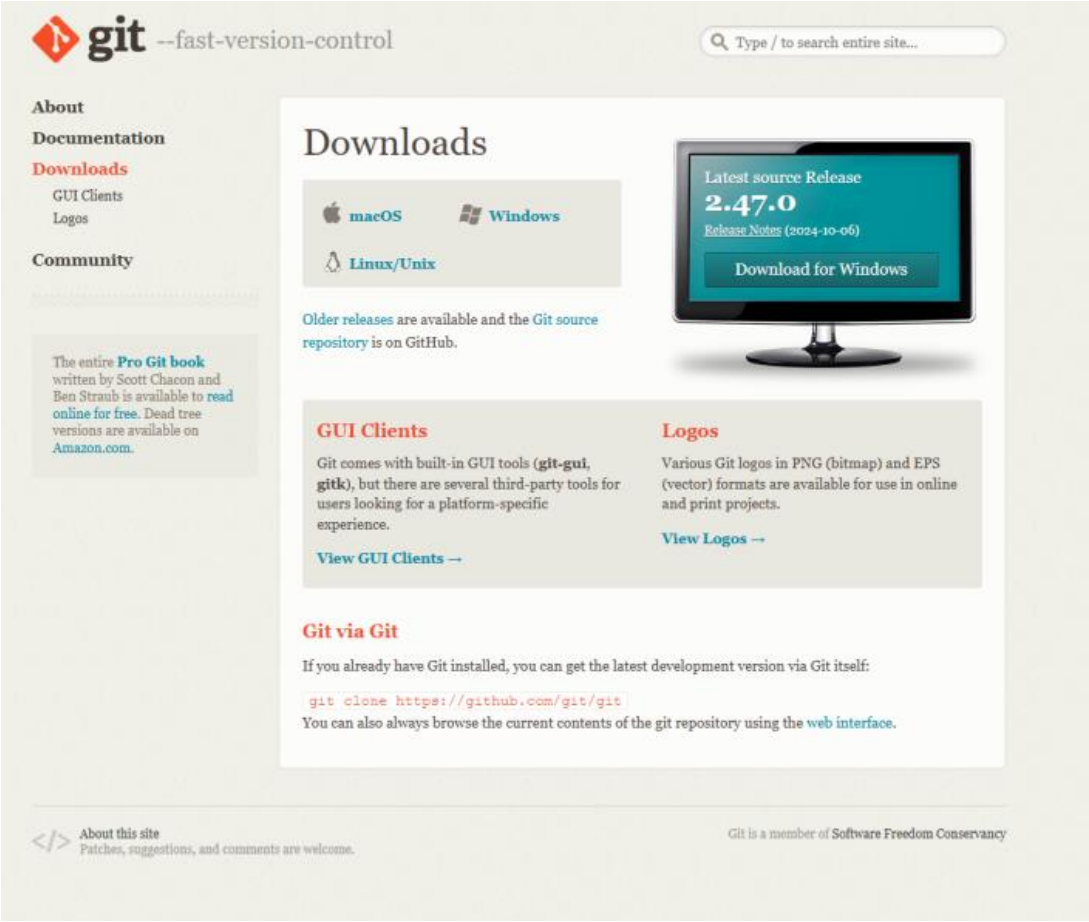
Task Manager

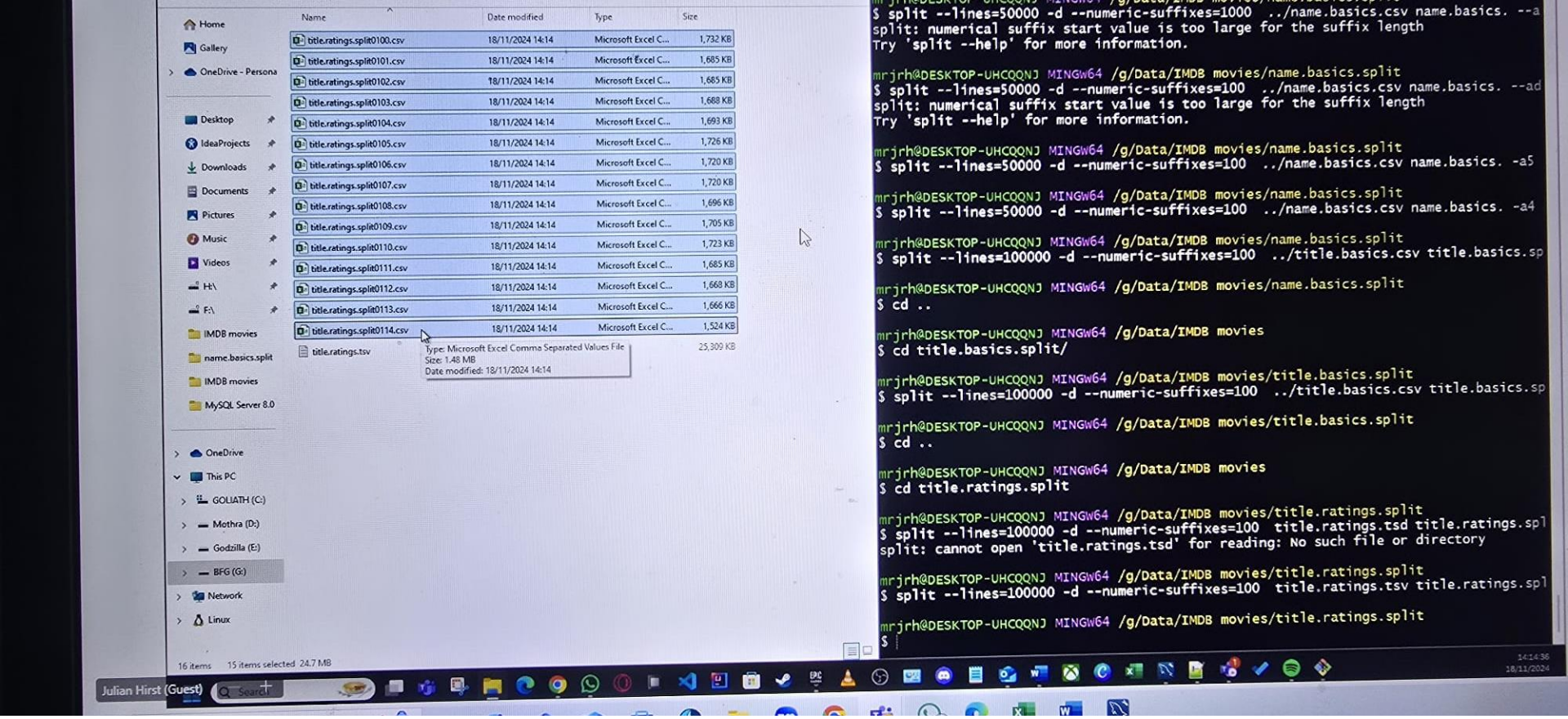
- The Task Manager helps monitor resource usage.
- mySQL workbench will still lose contact with the server on extremely long tasks.
- In this case, 'mysqld.exe' in task manager indicates MySQL is still working on something.

Name	Status	CPU	Memory	Disk	Network
MySQL Workbench		0%	120.4 MB	0.1 MB/s	0 Mbps
mysqld.exe (3)		11.4%	413.7 MB	52.3 MB/s	0 Mbps

Downloading split command

- The GIT installer includes a suite of programs including "split".
- "split" allowed us to cut datafiles into chunks of 100,000 records for easier processing.





Splitting files

- • Split used lit command to split files into smaller chunks of 100,000 lines/records.
- $ split --lines=100000 -d --numeric-suffixes=100 title.ratings.tsv title.ratings.split -a4 --additional-suffix=.csv

Excel to make the required LOAD DATA commands for each of the split files

="LOAD DATA INFILE " & D937 & "REPLACE INTO TABLE "&F937&" FIELDS TERMINATED BY " &"'\t'" &" LINES TERMINATED BY "& "'\n'" &" ("& E937 &"); Commit; "

E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1035.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1036.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1037.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1038.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1039.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1040.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1041.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1042.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1043.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1044.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1045.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1046.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1047.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOA															
eid,ordering, nameid, category, roles, characters	title_principals	LOA	DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1049.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED B								
eid,ordering, nameid, category, roles, characters	title_principals	LOA															
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1051.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1052.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1053.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1054.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1055.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1056.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1057.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1058.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1059.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1060.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1061.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1062.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1063.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1064.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									
eid,ordering, nameid, category, roles, characters	title_principals	LOAD DATA INFILE	"G:/Data/IMDB movies/title.principals/title.principals1065.csv"	REPLACE	INTO TABLE	title_principals	FIELDS TERMINATED BY '\t'	LINES TERMINATED BY									

We tried stored procedures for this but couldn't get it to work so used excel.

We inserted the commands created by excel into mysql and then run the query - which could take up to 20 minutes

```
989 •  LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals1078.csv" INTO TABLE title_principals  FIELDS TERMINATED BY '\t' LINES TERMINATED BY '\n' (titleid,o
990 •  LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals1079.csv" INTO TABLE title_principals  FIELDS TERMINATED BY '\t' LINES TERMINATED BY '\n' (titleid,o
991 •  LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals1080.csv" INTO TABLE title_principals  FIELDS TERMINATED BY '\t' LINES TERMINATED BY '\n' (titleid,o
992 •  LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals1081.csv" INTO TABLE title_principals  FIELDS TERMINATED BY '\t' LINES TERMINATED BY '\n' (titleid,o
993 •  LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals1082.csv" INTO TABLE title_principals  FIELDS TERMINATED BY '\t' LINES TERMINATED BY '\n' (titleid,o
994 •  LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals1083.csv" INTO TABLE title_principals  FIELDS TERMINATED BY '\t' LINES TERMINATED BY '\n' (titleid,o
995 •  LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals1084.csv" INTO TABLE title_principals  FIELDS TERMINATED BY '\t' LINES TERMINATED BY '\n' (titleid,o
996 •  LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals1085.csv" INTO TABLE title_principals  FIELDS TERMINATED BY '\t' LINES TERMINATED BY '\n' (titleid,o
997 •  LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals1086.csv" INTO TABLE title_principals  FIELDS TERMINATED BY '\t' LINES TERMINATED BY '\n' (titleid,o
998 •  LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals1087.csv" INTO TABLE title_principals  FIELDS TERMINATED BY '\t' LINES TERMINATED BY '\n' (titleid,o
999 •  LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals1088.csv" INTO TABLE title_principals  FIELDS TERMINATED BY '\t' LINES TERMINATED BY '\n' (titleid,o

1001 •  Commit;
1002
1003 •  SET FOREIGN_KEY_CHECKS = 1;
1004 •  SET UNIQUE_CHECKS = 1;
1005
```

Output

Action Output

#	Time	Action	Message
1673	14:04:01	Commit	0 row(s) affected
1674	14:04:01	LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals0636.csv" INTO TABLE title_principals FIELDS TERMINA...	100000 row(s) affected Records: 100000 Deleted: 0 Skipped: 0 Warnings: 0
1675	14:04:03	Commit	0 row(s) affected
1676	14:04:03	LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals0637.csv" INTO TABLE title_principals FIELDS TERMINA...	100000 row(s) affected Records: 100000 Deleted: 0 Skipped: 0 Warnings: 0
1677	14:04:04	Commit	0 row(s) affected
1678	14:04:04	LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals0638.csv" INTO TABLE title_principals FIELDS TERMINA...	100000 row(s) affected Records: 100000 Deleted: 0 Skipped: 0 Warnings: 0
1679	14:04:05	Commit	0 row(s) affected
1680	14:04:05	LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals0639.csv" INTO TABLE title_principals FIELDS TERMINA...	100000 row(s) affected Records: 100000 Deleted: 0 Skipped: 0 Warnings: 0
1681	14:04:06	Commit	0 row(s) affected
1682	14:04:06	LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals0640.csv" INTO TABLE title_principals FIELDS TERMINA...	100000 row(s) affected Records: 100000 Deleted: 0 Skipped: 0 Warnings: 0
1683	14:04:07	Commit	0 row(s) affected
1684	14:04:07	LOAD DATA INFILE "G:/Data/IMDB movies/title.principals/title.principals0641.csv" INTO TABLE title_principals FIELDS TERMINA...	Running

Further Work

title.akas.tsv.gz - Contai

- titleId (string) - a tconst,
- ordering (integer) – a nu
- title (string) – the localiz
- region (string) - the regi
- language (string) - the la
- types (array) - Enumera title. One or more of the fol "festival", "tv", "video", "wo values may be added in the
- attributes (array) - Addit title, not enumerated.
- isOriginalTitle (boolean)

title.basics.tsv.gz - Co

- tconst (string) - alph
- titleType (string) – th tvseries, tvepisode, vid
- primaryTitle (string) the filmmakers on prom
- originalTitle (string)
- isAdult (boolean) - 0
- startYear (YYYY) – re case of TV Series, it is t
- endYear (YYYY) – TV
- runtimeMinutes – pri
- genres (string array) the title.

title.principals.tsv.gz -

- tconst (string) - alpha
- ordering (integer) – a titleId.
- nconst (string) - alph
- category (string) - th
- job (string) - the spec
- characters (string) - t else.

title.ratings.tsv.gz – C

- tconst (string) - alpha
- averageRating – weig
- numVotes - number

name.basics.tsv.gz – Contains the

- nconst (string) - alphanumeric unic
- primaryName (string)– name by wh
- birthYear – in YYYY format.
- deathYear – in YYYY format if appli
- primaryProfession (array of strings person.
- knownForTitles (array of tconsts) –

Rationalising title_basics and title_ratings to titles_final

```
CREATE TABLE titles_final AS
SELECT
    t.titleid,
    t.titleType,
    t.primaryTitle,
    t.originalTitle,
    t.startYear,
    t.endYear,
    t.runtimeMinutes,
    tr.averageRating,
    tr.numVotes
FROM title_basics t
JOIN title_ratings tr
ON t.titleid = tr.titleid;
```

We dropped "genres" from "title_basics" as this would eventually be placed in a separate table "title_genres"

Testing

```
SELECT * FROM title_basics tb
JOIN title_ratings tr ON tb.titleid = tr.titleid
WHERE tb.startYear = 1983 AND tr.averageRating >= 8.3 and titleType="movie"
and numVotes>10000;
```

titleid	titleType	primaryTitle
tt0085743	movie	Jaane Bhi Do Yaaro
tt0086190	movie	Star Wars: Episode VI - Return of the Jedi
tt0086250	movie	Scarface

Checking results

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

titleid	primaryTitle	startYear	averageRating	numVotes
tt0252597	Feyzo, the Polite One	1978	8.7	17637
tt0765833	Happy Days	1978	8.7	12152
tt0078748	Alien	1979	8.5	986097
tt0079221	Gol Maal	1979	8.5	20372
tt0076276	Who's Singin' Over There?	1980	8.7	16917
tt0080684	Star Wars: Episode V - The Empir...	1980	8.7	1406556
tt0082096	Das Boot	1981	8.4	268841
tt0082971	Raiders of the Lost Ark	1981	8.4	1052154
tt0084302	The Marathon Family	1982	8.8	17251
tt0085743	Jaane Bhi Do Yaaro	1983	8.3	14924
tt0086190	Star Wars: Episode VI - Return o...	1983	8.3	1140825
tt0086250	Scarface	1983	8.3	937426
tt0086935	Balkan Spy	1984	8.8	12647
tt0090015	Shoah	1985	8.7	10780
tt0201368	The Broken Landlord	1985	8.7	16669
tt0090605	Aliens	1986	8.4	788229
tt0093603	Nayakan	1987	8.7	25450

With rationalised "titles_final" table, best movies by year

```
SELECT tb.titleid, tb.primaryTitle, tb.startYear, tr.averageRating, tr.numVotes
FROM title_basics tb
JOIN title_ratings tr ON tb.titleid = tr.titleid
WHERE tb.titleType = 'movie'
  AND tr.numVotes > 1000
  AND tr.averageRating = (
      SELECT MAX(tr2.averageRating)
      FROM title_ratings tr2
      JOIN title_basics tb2 ON tr2.titleid = tb2.titleid
      WHERE tb2.startYear = tb.startYear
  )
ORDER BY tb.startYear, tr.averageRating DESC;
```

One million million queries!

EXECUTION TIME ???

Estimate execution time at 1 second per sweep of the table

```
SELECT tb.titleid, tb.primaryTitle, tb.startYear, tr.averageRating, tr.numVotes
FROM title_basics tb
JOIN title_ratings tr ON tb.titleid = tr.titleid
WHERE tb.titleType = 'movie'
  AND tr.numVotes > 1000
  AND tr.averageRating = (
      SELECT MAX(tr2.averageRating)
      FROM title_ratings tr2
      JOIN title_basics tb2 ON tr2.titleid = tb2.titleid
      WHERE tb2.startYear = tb.startYear
  )
ORDER BY tb.startYear, tr.averageRating DESC;
```

One million million queries!

1 million seconds = 11.5 days

Using "WITH" & original tables

```
WITH RankedMovies AS (
   SELECT tb.titleid, tb.primaryTitle, tb.startYear, tr.averageRating, tr.numVotes,
         RANK() OVER
               (PARTITION BY tb.startYear ORDER BY tr.averageRating DESC)
         AS ratingRank
   FROM title_basics tb
   JOIN title_ratings tr ON tb.titleid = tr.titleid
   WHERE tb.titleType = 'movie'
     AND tr.numVotes > 10000
)
SELECT titleid, primaryTitle, startYear, averageRating, numVotes
FROM RankedMovies
WHERE ratingRank = 1
ORDER BY startYear, averageRating DESC;
```

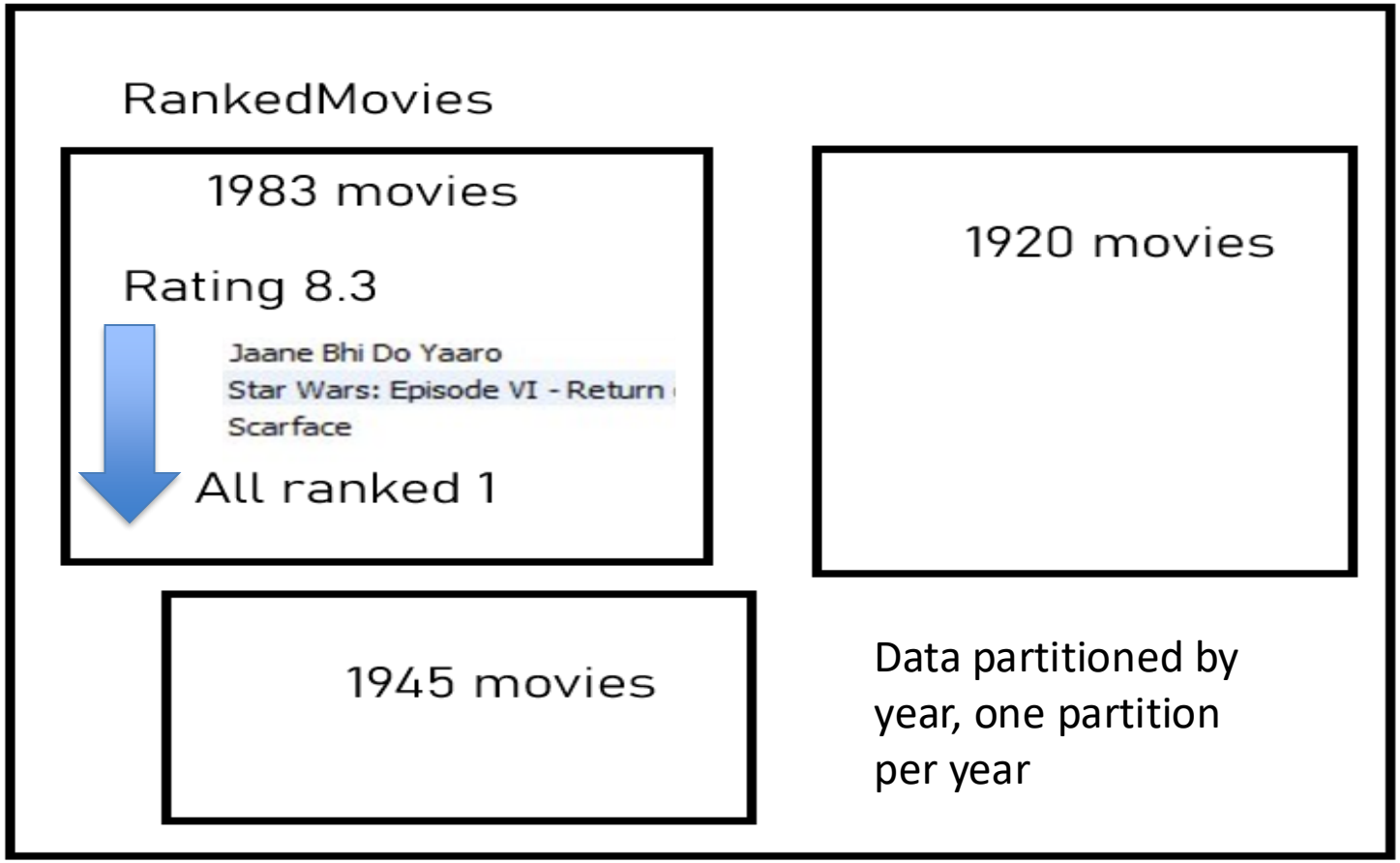
EXECUTION TIME 2-3 SECONDS

WITH + with tableFinal

```
WITH RankedMovies AS (
   SELECT tf.titleid, tf.primaryTitle, tf.startYear, tf.averageRating, tf.numVotes,
         RANK() OVER (PARTITION BY tf.startYear
         ORDER BY tf.averageRating DESC) AS ratingRank
   FROM titles_final tf
   WHERE tf.titleType = 'movie'
     AND tf.numVotes > 10000
)
SELECT titleid, primaryTitle, startYear, averageRating, numVotes
FROM RankedMovies
WHERE ratingRank = 1
ORDER BY startYear, averageRating DESC;
```

EXECUTION TIME 1-2 SECONDS

Window Functions



Indexing Table

Index Name	Type
PRIMARY	PRIMARY
titleid_UNIQUE	UNIQUE
yeari	INDEX
rating	INDEX
yearRating	INDEX

Index Columns

Column	#	Order	Length
titleid		ASC	
titleType		ASC	
primaryTitle		ASC	
originalTitle		ASC	
startYear	1	ASC	
endYear		ASC	
runtimeMinutes		ASC	
averageRating	2	ASC	
numVotes		ASC	

SQL using index

```
SELECT
   tf.titleid,
   tf.primaryTitle,
   tf.startYear,
   tf.averageRating,
   tf.numVotes
FROM titles_final tf USE INDEX (yearRating)
WHERE tf.titleType = 'movie'
   AND tf.numVotes > 1000
   AND tf.averageRating = (
      SELECT MAX(tf2.averageRating)
      FROM titles_final tf2 USE INDEX (yearRating)
      WHERE tf2.startYear = tf.startYear
   )
ORDER BY tf.startYear, tf.averageRating DESC;
```

After Indexing 353 seconds

SQL using index

```
SELECT tb.titleid, tb.primaryTitle, tb.startYear, tr.averageRating, tr.numVotes
FROM title_basics tb
JOIN title_ratings tr ON tb.titleid = tr.titleid
WHERE tb.titleType = 'movie'
  AND tr.numVotes > 1000
  AND tr.averageRating = (
      SELECT MAX(tr2.averageRating)
      FROM title_ratings tr2
      JOIN title_basics tb2 ON tr2.titleid = tb2.titleid
      WHERE tb2.startYear = tb.startYear
  )
ORDER BY tb.startYear, tr.averageRating DESC;
```

After Indexing ??? seconds